

Daniel Gray

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Senior Data Engineer

Professional Summary

Strategic and technically rigorous **Senior Data Engineer** with **10+** years of experience designing, building, and optimizing scalable data platforms across SaaS, Finance, and Government domains. Expertise in **Snowflake, Python, Spark**, and SQL, with deep experience in **data lakehouse** design, real-time data processing, and ML-ready pipelines. Adept at building and optimizing **Snowflake-based** data architectures and managing large-scale data **warehouses**. Skilled in orchestrating complex workflows using **Airflow** and deploying cloud-native infrastructure on **AWS, GCP**, and **Azure**. Proven ability to deliver secure, high-performance data engineering systems and enable actionable analytics with **Snowflake** and **Power BI**. Strong leadership and mentoring skills, with a focus on fostering collaboration and driving business insights.

Skills

- **Languages:** Python 3.11+, SQL, Scala, Bash
- **Data Engineering:** Apache Spark, PySpark, Pandas, Dask, Databricks, Airflow, dbt, Kafka, Flink, Beam
- **Storage & Modeling:** Snowflake, Delta Lake, Iceberg, Parquet, ORC, Avro, Data Vault, Star Schema,
- **Databases:** Snowflake, PostgreSQL, MySQL, BigQuery, Redshift, MongoDB
- **Cloud Platforms:** AWS (Glue, EMR, S3, Lambda, RDS), GCP (Dataflow, BigQuery, Pub/Sub), Azure (Data Factory, Synapse, Blob Storage)
- **Orchestration:** Airflow, Prefect, Dagster, Luigi
- **Streaming:** Kafka, Kinesis, Pub/Sub, Spark Structured Streaming
- **DevOps & CI/CD:** Docker, Kubernetes, Terraform, GitHub Actions, Jenkins
- **Security:** IAM, RBAC, Data Lineage, Encryption, GDPR, HIPAA
- **Monitoring:** Prometheus, Grafana, ELK, CloudWatch, OpenTelemetry
- **AI/ML Integration:** Feature Stores, MLflow, TensorFlow, scikit-learn, SageMaker Pipelines
- **Business Intelligence:** Power BI, Tableau, Looker Studio
- **Soft Skills:** Technical Leadership, Mentorship, Agile, Documentation

Work Experience

Senior Data Engineer / Pearl Health

Hollywood, FL / Jan 2022 – Present

- Designed and optimized data models in **Snowflake** for multi-terabyte datasets at **Pearl Health**, improving query performance by 40% and enabling real-time, accurate analytics for cross-functional teams, particularly for healthcare data insights.
- Developed **Snowflake-based** data **warehousing** solutions at **Pearl Health**, integrating data from **AWS S3**, **EMR**, and **FHIR** APIs to centralize healthcare-related data, supporting reporting and decision-making with real-time insights for business stakeholders.
- Architected and built distributed data pipelines using **PySpark**, **Airflow**, **Delta Lake**, and **Snowflake**, processing over 10TB of data daily across **AWS Glue**, **S3**, and Snowflake, ensuring high scalability and fault tolerance for healthcare data workflows.
- Implemented advanced **Snowflake** features like automatic clustering, **zero-copy** cloning, and data sharing to optimize query performance and manage data lifecycle in a regulated healthcare environment.
- Orchestrated ETL workflows with **Airflow**, automating data extraction, transformation, and loading processes for **Snowflake** at **Pearl Health**, enhancing data pipeline reliability and minimizing job failure rates while maintaining HIPAA-compliant data security.
- Mentored junior engineers at **Pearl Health** on optimizing **Snowflake** performance, focusing on data partitioning, clustering keys, and cost-optimization strategies to ensure efficient data querying and reduce cloud resource expenses.
- Designed and implemented data access control policies in **Snowflake** using role-based access control (**RBAC**) to meet **SOC 2** and **HIPAA** compliance requirements for sensitive healthcare data access at **Pearl Health**.
- Enabled seamless data integration between **Snowflake** and **AWS** services like **S3**, **Lambda**, and **Redshift** to streamline data flow and enhance analytics capabilities across healthcare platforms at **Pearl Health**.
- Implemented robust **data quality frameworks** in **Snowflake** using tools like **Great Expectations**, **dbt**, and real-time monitoring with **Prometheus** and **Grafana**, ensuring data integrity for critical healthcare analytics at **Pearl Health**.

Senior Data Engineer / eBay

San Jose, CA / May 2020 – Dec 2021

- Led development of scalable analytics platforms using **Apache Beam**, **Dataflow**, **BigQuery**, and **Snowflake**, powering analytics and ML systems that drove business insights and decision-making at **eBay**.
- Built and optimized streaming systems using **Pub/Sub**, **Kafka**, **Beam**, and **Snowflake**, enabling low-latency ingestion for real-time product and transactional data analysis across eBay's global marketplace.
- Designed and maintained **data marts** in **Snowflake**, along with **dbt** and Looker, enabling self-service analytics for eBay's product, marketing, and operations teams to analyze customer behavior, sales trends, and inventory performance.

- Developed **ETL** pipelines using **Airflow**, **Python**, and **Snowflake**, integrating internal systems (e.g., inventory management, pricing engines) to enable seamless data flow and analytics, optimizing data storage in Snowflake.
- Implemented data governance practices using **Snowflake** Data Catalog, IAM, and automated sensitive-data classification to ensure compliance with eBay's data policies and enhance data security across platforms.
- Built **ML pipelines in Snowflake** with integration to Vertex AI, TensorFlow, and MLflow, enabling eBay's data science team to deploy and manage machine learning models for pricing optimization, recommendation systems, and fraud detection.
- Created internal data profiling tools using **Streamlit**, **Flask**, and **Snowflake**, allowing teams to track data quality, lineage, and ensure data integrity across eBay's ecosystem.
- Automated infrastructure provisioning using **Terraform**, deploying Snowflake solutions along with **GKE**, **Cloud Run**, and **Cloud Composer**, ensuring efficient scaling of data pipelines and smooth integration of new services into eBay's cloud-native architecture.
- Optimized **workloads** in **Snowflake**, using **Query Optimization** techniques, BigQuery optimizers, and Spark UI to improve query performance, reduce costs, and ensure efficient processing of eBay's massive datasets.
- Led architecture reviews and authored technical design documents for **Snowflake**-based systems, aligning eBay's data strategy with business goals, including scaling real-time data pipelines and supporting e-commerce insights.
- Mentored engineers on best practices for Snowflake data modeling, **streaming data integration**, and **ML Ops**, improving team expertise and driving efficient data engineering workflows.
- Delivered government-grade **data modules** with **Snowflake** data masking, audit trails, and compliance logging, ensuring eBay's privacy and security standards for handling sensitive customer and transaction data.
- Created **automated tests for Snowflake pipelines** using pytest, Great Expectations, and Postman, ensuring high-quality, reliable, and production-ready data processing workflows.
- Authored and maintained comprehensive documentation for Snowflake-based architectures, ensuring clarity for internal stakeholders and ease of maintenance using Markdown, Sphinx, and GitHub Wiki.

Lead Data Engineer / Devoted Health

Waltham, MA / July 2018 – Apr 2020

- Led development of data pipelines for healthcare and insurance clients using Python, SQLAlchemy, and Airflow, integrating key systems for tax and payroll processing to ensure seamless data flow and compliance.
- Built and optimized ingestion pipelines using Flask, cron, and PostgreSQL, connecting with third-party APIs for real-time data synchronization across healthcare and insurance systems.
- Designed and implemented schemas for SQL Server and AWS Redshift, optimizing indexing, partitioning, and stored procedures for improved data access and query performance in large-scale healthcare data.

- Developed custom reconciliation tools with pandas, Jupyter, and custom validation rules to ensure data consistency across multiple financial and healthcare systems.
- Delivered secure modules adhering to RBAC, encryption, and audit logging to meet IRS, SOC2, and HIPAA standards for sensitive data handling in healthcare and insurance applications.
- Refactored legacy ETL processes into modular Python components, following PEP8 guidelines, integrating type hints, and writing unit tests to ensure maintainability and testability of code in a fast-paced startup environment.
- Automated CI/CD deployments using TeamCity, Docker, and Fabric, enabling streamlined deployment of data pipelines to both Azure and AWS environments, reducing deployment errors and improving operational efficiency.
- Built compliance dashboards using Dash, Plotly, and SSRS, enabling stakeholders to visualize healthcare claims data, tax calculations, and payroll compliance metrics in real-time.
- Participated in Agile ceremonies, including sprint planning, QA collaboration, and code reviews, fostering collaboration within a cross-functional team of data engineers and analysts in a rapidly growing healthcare startup.
- Authored comprehensive internal documentation using Markdown and Confluence, ensuring all data processes and systems were well-documented for current and future team members.
- Designed data security strategies using AWS IAM, Azure Active Directory, and Lake Formation to protect sensitive healthcare data, ensuring compliance with HIPAA, SOC2, and other industry standards.
- Led architecture reviews and provided mentorship to junior engineers on data modeling, cloud integration, and ETL best practices, driving high-quality data pipeline development in a startup environment.
- Integrated AWS and Azure cloud services, including AWS Lambda, Azure Data Factory, and Azure Synapse Analytics, to implement scalable, cloud-native data pipelines for handling healthcare claims, payroll, and tax data.
- Collaborated with the product and engineering teams to build innovative solutions that streamlined claims management, benefits administration, and payroll processing, improving operational efficiency and reducing errors.

Data Engineer / Capgemini

New York, NY | Nov 2016 – Jun 2018

- Developed scalable ETL pipelines for various consulting clients using AWS Glue, GCP Dataflow, and Azure Data Factory, ensuring seamless data ingestion and transformation across cloud platforms.
- Built data lakes on AWS S3, GCP Cloud Storage, and Azure Blob Storage to centralize client data, enabling real-time analytics and machine learning use cases.
- Designed and optimized SQL databases and data warehouses on AWS Redshift, Google BigQuery, and Azure Synapse Analytics, improving data querying speed and analytics performance.
- Automated cloud data processing workflows using AWS Lambda, Google Cloud Functions, and Azure Functions, enabling cost-efficient serverless operations for clients.
- Led cloud data migration projects, moving large-scale datasets from on-premises systems to AWS, GCP, and Azure, ensuring smooth transitions and minimal downtime for enterprise clients.

- Implemented data security and access control strategies using AWS IAM, GCP IAM, and Azure AD, ensuring compliance with client-specific security requirements (e.g., GDPR, SOC2).
- Designed and deployed real-time streaming systems using AWS Kinesis, GCP Pub/Sub, and Azure Event Hubs, enabling clients to ingest and analyze high-velocity data streams with low latency.
- Integrated third-party APIs and external data sources into clients' data ecosystems using Python, AWS API Gateway, Google Cloud Endpoints, and Azure API Management.
- Created data pipelines for batch and streaming data using Apache Kafka, AWS Kinesis, and GCP Dataflow, ensuring timely data delivery for business-critical applications.
- Built data models using dbt, Snowflake, and Google BigQuery for various clients, enabling efficient analytics and self-service reporting.
- Developed custom data validation tools with Great Expectations and pandas, ensuring high-quality data pipelines and reducing errors in data transformation and reporting.
- Collaborated with cross-functional teams to implement data governance frameworks using AWS Lake Formation, GCP Data Catalog, and Azure Purview for metadata management and compliance.
- Designed and implemented data profiling and quality monitoring tools with AWS CloudWatch, GCP Stackdriver, and Azure Monitor, providing real-time alerts and metrics for clients' data health.
- Implemented cost optimization strategies across AWS, GCP, and Azure by optimizing cloud resource allocation and utilizing services like AWS Spot Instances, GCP Preemptible VMs, and Azure Reserved Instances.
- Developed dashboards and reporting solutions using Power BI, Tableau, and Looker, integrating data from AWS Redshift, GCP BigQuery, and Azure Synapse to provide clients with actionable insights.
- Mentored junior engineers in cloud data best practices, ETL pipeline design, and data integration, ensuring consistent high-quality deliveries for consulting clients.
- Collaborated with clients to design custom machine learning pipelines using AWS SageMaker, GCP AI Platform, and Azure Machine Learning, helping clients unlock insights from their data.
- Designed and implemented CI/CD pipelines for cloud-based data services using Jenkins, GitHub Actions, and Azure DevOps, ensuring seamless updates and deployments.
- Conducted architecture reviews and technical design sessions with clients to propose optimal solutions on AWS, GCP, and Azure based on their specific business needs.

Python Developer / University of Florida

Gainesville, FL | May 2015 – Oct 2016

- Worked as a Python developer while pursuing a Master of Science in Computer Science at the University of Florida, applying academic knowledge to real-world projects.
- Developed data-driven web applications using Python, Flask, and SQLAlchemy for managing student records, course registrations, and academic data.
- Created RESTful APIs in Python with Flask to integrate with frontend applications and deliver dynamic content for student-facing modules and faculty dashboards.
- Built interactive user interfaces using React and Angular, focusing on responsive design and real-time data binding to enhance the user experience for students and faculty.

- Designed and implemented course registration systems with Python, using PostgreSQL for database management and enabling students to register for courses and track academic progress.
- Developed real-time data visualization tools using React and D3.js, helping academic departments track student performance, course enrollment trends, and academic calendar data.
- Participated in Agile development cycles, including sprint planning, daily standups, and code reviews, ensuring timely delivery of project milestones while learning software engineering best practices.
- Integrated third-party APIs for authentication and user management using OAuth2 and JWT, allowing secure login and role-based access to university systems.
- Implemented data processing pipelines in Python, automating data collection and transformation tasks for research projects and academic analyses.
- Led the development of academic tools, including student progress trackers and classroom management systems, using Angular for frontend components and Flask for backend processing.
- Utilized Git and GitHub for version control, ensuring code collaboration with peers and proper documentation throughout the development lifecycle.
- Developed modular, reusable components in React and Angular, following best practices and principles such as component-based architecture to ensure scalability and ease of maintenance.
- Participated in cross-functional collaboration with database administrators and faculty to ensure academic data was accurately represented and efficiently queried.
- Implemented unit and integration tests using pytest and Jest to ensure the reliability and functionality of Python backends and JavaScript frontend components.
- Enhanced website performance using lazy loading, async/await, and Webpack bundling, reducing loading times and optimizing frontend responsiveness.
- Authored detailed project documentation in Markdown and contributed to knowledge sharing on GitHub Wiki, enabling future students and developers to understand and build on existing systems.

Education

University of Florida

 Gainesville, FL

 Master of Science in Data Science

 May 2015 – Oct 2016

University of Florida

 Gainesville, FL

 Bachelor of Science in Information Technology

 Aug 2011 – May 2015